Alessandro Pani Patiño

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Summary

Computer Science student at the University of Costa Rica who builds and ships production software. Replaced a paper-based process at UCR with a web app used by 50+ staff, and has built and supported websites for 10+ freelance clients since 2022. Experienced in Agile (Scrum), multidisciplinary teams. Available full-time in 2027.

Education

University of Costa Rica | B.S. Computer Science (Software Eng.) *Coursework ends Dec 2027*

Experience

Research & Development Assistant | Ensamblaje Piñero, UCR *Jun 2026 - Present*

- Sole developer on a UCR linguistics research project: built Python pipelines that collected and segmented 7,063 Legislative Assembly transcripts (~17 GB) into a traceable corpus, plus user guides for the team.
- Building a benchmark of Spanish part-of-speech taggers, from pretrained pipelines (Stanza, spaCy) to a fine-tuned transformer language model (BETO), with a strict EAGLES evaluator and 67 automated tests.
- Prepared SLURM jobs to fine-tune BETO on NVIDIA A100 GPUs in UCR's HPC cluster.

Full-Stack Developer | SIGECAT, UCR Vice-Rectorate of Administration *Jan 2026 - Jun 2026*

- Replaced paper-based work-schedule and shift declarations with a production web application used by 50+ university staff; built and integrated the React + TypeScript frontend, PHP REST API and Oracle data layer.
- Cut slow Oracle query times from about 4 s to under 1 s; verified the API with endpoint and concurrency tests.
- Secured the API with token authentication and refresh-token rotation, bcrypt, per-IP rate limiting, account lockout and prepared statements; shipped with Docker and CI (lint, build, PHPStan) on Oracle Cloud.

Freelance Web Developer *2022 - Present*

- Built and maintained websites for 10+ clients in legal services, hospitality and e-commerce, owning the full development lifecycle: requirements, development, testing, deployment and SEO.
- Troubleshoot client issues, configure email, DNS, SSL and hosting, and train clients to edit their content.

Software Engineer | deferia.cr, UCR and FAO *Aug 2023 - May 2025*

- Joint UCR and UN FAO project: turned four years of CNP price bulletins into a clean 9,184-record dataset across 56 products (Django, PostgreSQL) and built interactive price charts (React, Plotly).

Projects

Mods | AR Custom Cap Store, UCR Software Quality course *mods-cr.vercel.app*

- Built a web store where customers design a trucker cap and try it on in 3D with their phone camera; face tracking runs on-device, so no image leaves the phone (TypeScript, Three.js, MediaPipe).
- Tested with 146 unit tests (Vitest) and Playwright end-to-end tests (privacy, offline, checkout); designed a GQM plan of 15 code and process quality metrics and 16 V&V activities.

ECCI QR Access Control, UCR *Code on GitHub*

- Interviewed faculty, security staff and students, then built a QR sign-in demo with role-based access.

Museo del Espacio Matemático | 30+ WebGL simulations, hand tracking *Live demo*

Skills

Languages: Python, TypeScript/JavaScript, PHP, SQL (Oracle, PostgreSQL, SQLite)
Web: React, Astro, Django, REST APIs
Data and ML: pandas, scikit-learn, Power BI, PyTorch, Hugging Face Transformers, spaCy, Stanza
Tools: Git, Docker, Linux/Bash, SLURM (HPC), pytest, CI pipelines, Jira, Agile/Scrum, Oracle Cloud
Spoken: Spanish and Italian (native), English (C1, professional)